

Scalability & Future Plan

Date	
Team ID	
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset with only 2200 records and 22 crop types	Limited crop diversity; model may not generalize well to regions with different crops or soil types not represented in the dataset	High
2	No user authentication or multi-tenant support	Cannot personalize recommendations per user/farm; no history tracking per farmer or location	Medium
3	Single-threaded Flask server (no async/queue system)	Under heavy concurrent load, requests may queue up; chart generation blocks API responses during first load	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask process; handles ~50 concurrent requests	Deploy with multiple gunicorn workers (already in Procfile). Add Redis-based caching for predictions. Auto-scale on Render with multiple instances.
2	Data Storage	Static CSV (Crop_recommendation.csv) loaded in memory	Migrate to PostgreSQL/SQLite database. Add API endpoints to insert new crop data. Admin dashboard for dataset management. Cloud storage (AWS S3) for chart images.

3	Performance	Charts regenerated on each server restart; model loaded from disk on every request	Pre-generate charts and store on CDN. Implement lazy-loading for dashboard sections. Model warm-up on deploy. Add Redis caching for frequent predictions.
4	Security	No authentication; open CORS (*); no rate limiting	Add JWT-based authentication for multi-user support. Restrict CORS to specific frontend domains. Implement rate limiting (Flask-Limiter). Input sanitization and CSRF protection.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2 (Short-term)	- User authentication (JWT) and multi-farm profiles - PostgreSQL database for dynamic data - Per-user prediction history tracking - Input validation with range sliders on frontend - Dark mode UI option	Q3 2026	Personalized experience for farmers. Persistent storage enables historical analysis of recommendations over time.
Phase 3 (Medium-term)	- Real-time weather API integration (OpenWeatherMap) - Fertilizer recommendation module - Crop disease detection using image classification (CNN) - Multi-language support (regional Indian languages) - Mobile-responsive PWA with offline support	Q1 2027	Reduces manual data entry by auto-filling weather data. Expands scope beyond crop selection to full farm advisory. Reaches non-English speaking farmers.
Phase 4 (Long-term)	- IoT sensor data integration (soil sensors, drones) - AI-powered yield prediction and pricing forecast - Government scheme and subsidy recommendation - Community features (farmer forum, expert Q&A) - Mobile app (Flutter/React Native)	Q3 2027	End-to-end precision agriculture platform. Connects farmers with real-time field data, market trends, and government support programs.